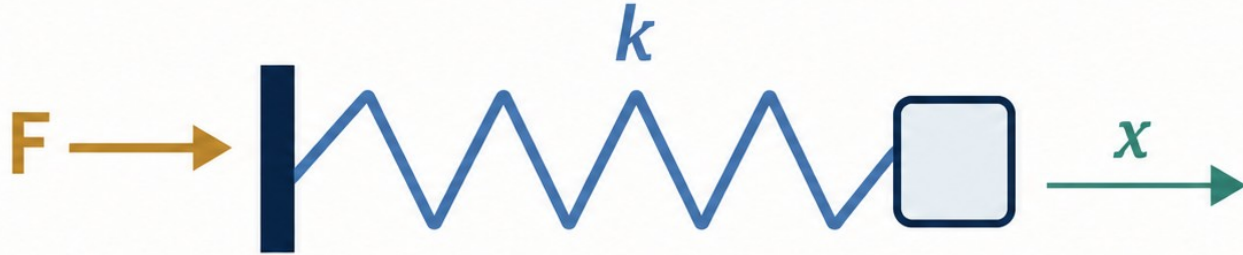


# Change one parameter, see the physics

---

Week 04 | Parameter sweeps and graph interpretation



$$x = F / k$$

# Predict before MATLAB

Model: ideal elastic spring (Hooke's law)

$$F = kx, \quad x = \frac{F}{k}$$

Same applied force  $F$

compare extension while  $k$  changes



$k = 50 \text{ N/m}$   
largest extension



$k = 100 \text{ N/m}$   
baseline



$k = 200 \text{ N/m}$   
smallest extension

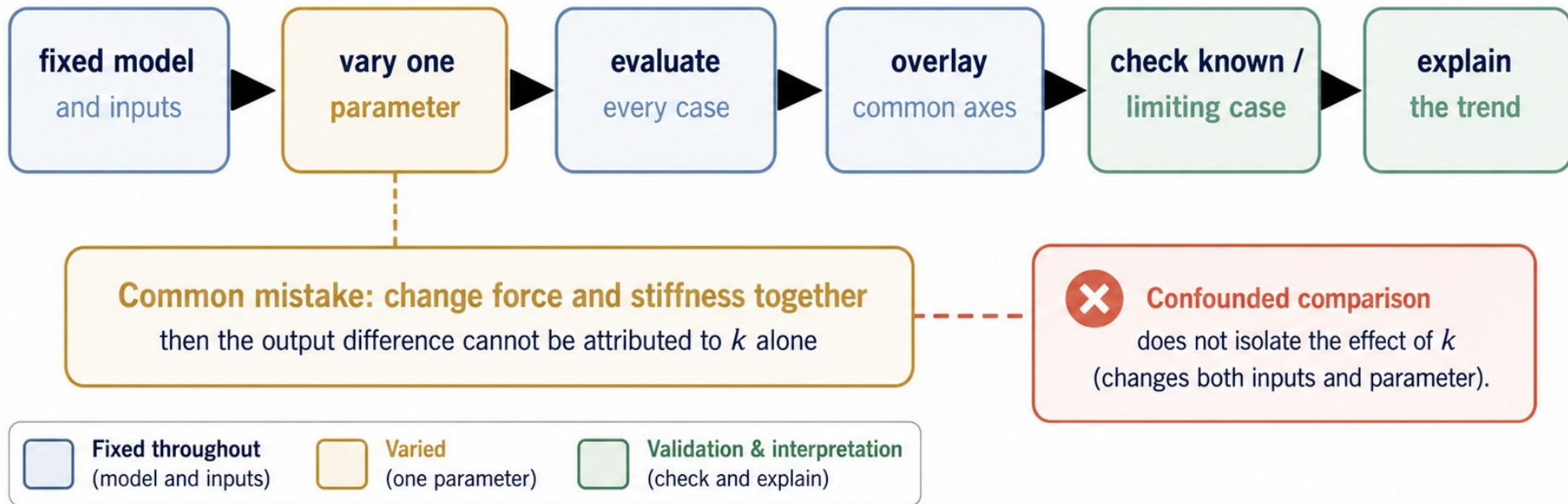


## Your predictions

- 1 Hooke's law gives  $x = F/k$ .
- 2 For the same positive force, **smaller  $k$  means larger extension.**
- 3 At  $F = 10 \text{ N}$ , predict **0.20 m**, **0.10 m**, and **0.05 m** for  $k = 50, 100$ , and  $200 \text{ N/m}$ .
- 4 At  $F = 0$ , every ideal-spring case should give  $x = 0$ .

# A sweep is a controlled experiment

- ✓ Vary one named parameter: spring stiffness  $k$ .
- ✓ Hold the force array, model, units, and all other assumptions fixed.
- ✓ A comparison that changes force and stiffness together is confounded.
- ✓ The fixed-input rule lets the output difference be attributed to  $k$ .





# Start with the model, variables, and units

- ✓ Use an ideal linear spring in its elastic range.
- ✓  $F = kx$ , so  $x = F/k$ .
- ✓  $F\_N$  is the common applied-force array in N.
- ✓  $k\_Npm$  is the swept stiffness in N/m; extension\_m is the output in m.

## Physical model: Ideal linear spring

$$F = kx, \quad x = \frac{F}{k}$$

$F$  = applied force (N)

$k$  = spring stiffness (N/m)

$x$  = spring extension (m)

## Variables and roles

| Variable  | Meaning / Role                                          | Unit |
|-----------|---------------------------------------------------------|------|
| $F\_N$    | Applied force<br>(common input<br>for all cases)        | N    |
| $k\_Npm$  | Spring stiffness<br>(swept parameter)                   | N/m  |
| $x = F/k$ | Spring extension<br>(output for each<br>stiffness case) | m    |

Baseline\_k\_Npm = 100 N/m (reference case)

## Dimensional check

$$\frac{\text{N}}{\text{N/m}} = \text{m}$$

Force (N) divided by  
stiffness (N/m) gives  
extension in metres (m).



Units are consistent.

# Plan the computation in plain language

|   |                                                                                   |                                               |                                                                                                                                        |
|---|-----------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1 |  | <b>State the model</b>                        | Use Hooke's law for an ideal spring: $F = kx$ and $x = F/k$ .                                                                          |
| 2 |  | <b>Choose swept parameter</b>                 | Sweep spring stiffness $k$ (N/m). Define cases: $k = [50 \quad 100 \quad 200]$ N/m.                                                    |
| 3 |  | <b>Fix all other inputs</b>                   | Force array $F_N = 0:0.5:10$ N, equation $x = F/k$ , units (N, m), assumptions, and any other inputs remain fixed.                     |
| 4 |  | <b>Evaluate the same model for every case</b> | For each $k$ case, compute extension $x = F/k$ at every force value.                                                                   |
| 5 |  | <b>Store results in a 3-by-21 arrangement</b> | One row per stiffness case; one column per force value. Rows correspond to $k = 50, 100, 200$ N/m.                                     |
| 6 |  | <b>Overlay on common axes with legend</b>     | Plot spring extension $x$ (m) versus applied force $F$ (N) for all cases on the same axes with labels and legend.                      |
| 7 |  | <b>Validate a known or limiting case</b>      | Check $F = 0$ gives $x = 0$ for every case. (Zero force $\rightarrow$ zero extension.)                                                 |
| 8 |  | <b>Explain the trend physically</b>           | Higher stiffness $k \rightarrow$ smaller extension $x$ for the same force $F$ . ( $x$ is inversely proportional to $k$ at fixed $F$ .) |

## One controlled change rule

This is a controlled computational experiment.

### Change (swept)

Spring stiffness  
 $k$  (N/m)



### Fixed

- ✓ Force array  $F_N = 0:0.5:10$  N
- ✓ Model  $x = F/k$
- ✓ Units (N, m)
- ✓ Assumptions
- ✓ Other inputs

Vary one parameter. Hold everything else constant.



Known (limiting) check:  $F = 0$  N  $\rightarrow x = 0$  m for all  $k$ .  
Any curve not passing through the origin is physically incorrect.



### Common defect to avoid (runnable but physically wrong):

Using  $x = F \times k$  instead of  $x = F/k$  gives larger  $x$  for larger  $k$ .  
Trend contradicts Hooke's law and physics.



# Store the cases explicitly

```
F_N = 0:0.5:10;  
% applied force array (N)  
  
k_Npm = [50 100 200];  
% spring stiffness values (N/m)  
  
baseline_k_Npm = 100;  
% baseline spring stiffness (N/m)
```

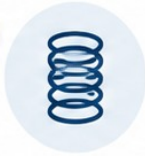
1



**Build one common force array.**

$F_N$  is the fixed input used for every stiffness case.

2



**Store the stiffness cases explicitly.**

$k_{Npm}$  lists the swept values 50, 100, and 200 N/m.

3



**Keep the baseline named so it can be traced later.**

$baseline\_k_{Npm} = 100$  N/m.

4



**Units in variable names help keep the physical meaning visible.**

N for Newtons, m for metres.

# Evaluate the same model for every case

- ✓ Preallocate one row for each stiffness value.
- ✓ Each loop pass changes only  $k_{\text{Npm}}(\text{case\_id})$ .
- ✓ Use element-wise division for  $x = F/k$ .
- ✓ Row  $\text{case\_id}$  maps to one stiffness case.

## MATLAB sweep loop

```
extension_m = zeros(length(k_Npm),length(F_N));  
for case_id = 1:length(k_Npm)  
    extension_m(case_id,:) = F_N./k_Npm(case_id);  
end
```

## Result matrix: extension\_m (size 3-by-21)

Each row = one stiffness case; each column = one applied force value

| Applied force, F_N (N) |        |       |        |     |      | $k_{\text{Npm}}(\text{case\_id})$<br>(N/m) |
|------------------------|--------|-------|--------|-----|------|--------------------------------------------|
| 0                      | 0.5    | 1     | 1.5    | ... | 10   |                                            |
| 0                      | 0.01   | 0.02  | 0.03   | ... | 0.20 | 50 N/m                                     |
| 0                      | 0.005  | 0.01  | 0.015  | ... | 0.10 | 100 N/m                                    |
| 0                      | 0.0025 | 0.005 | 0.0075 | ... | 0.05 | 200 N/m                                    |

21 applied-force values:  $F_N = 0:0.5:10$  N



## Known or limiting case check:

At  $F = 0$ , every row gives  $x = 0$ .

This confirms the model and computation are consistent.



**Interpretation:** Larger stiffness ( $k$ ) produces smaller extension ( $x$ ) for the same applied force ( $F$ ), matching Hooke's law  $x = F/k$ .

# Read one row before plotting

✓ At  $F = 10$  N, row 1 gives 0.20 m.

✓ Row 3 gives 0.05 m.

✓ The baseline row gives 0.10 m.

✓ The same-input ordering is  $x_{50} > x_{100} > x_{200}$ .

|     | $k$ (N/m) | $x$ at $F = 10$ N (m) | case             |
|-----|-----------|-----------------------|------------------|
| 1 → | 50        | 0.20                  | lower stiffness  |
| 2 → | 100       | 0.10                  | baseline         |
| 3 → | 200       | 0.05                  | higher stiffness |

## Same-input comparison at $F = 10$ N

All three values are for the same applied force. Read down the column to compare the effect of spring stiffness.

$x_{50} = 0.20$  m  
(row 1)

>

$x_{100} = 0.10$  m  
(baseline)

>

$x_{200} = 0.05$  m  
(row 3)

stiffness increases to the right → extension decreases



# Overlay every case on common axes

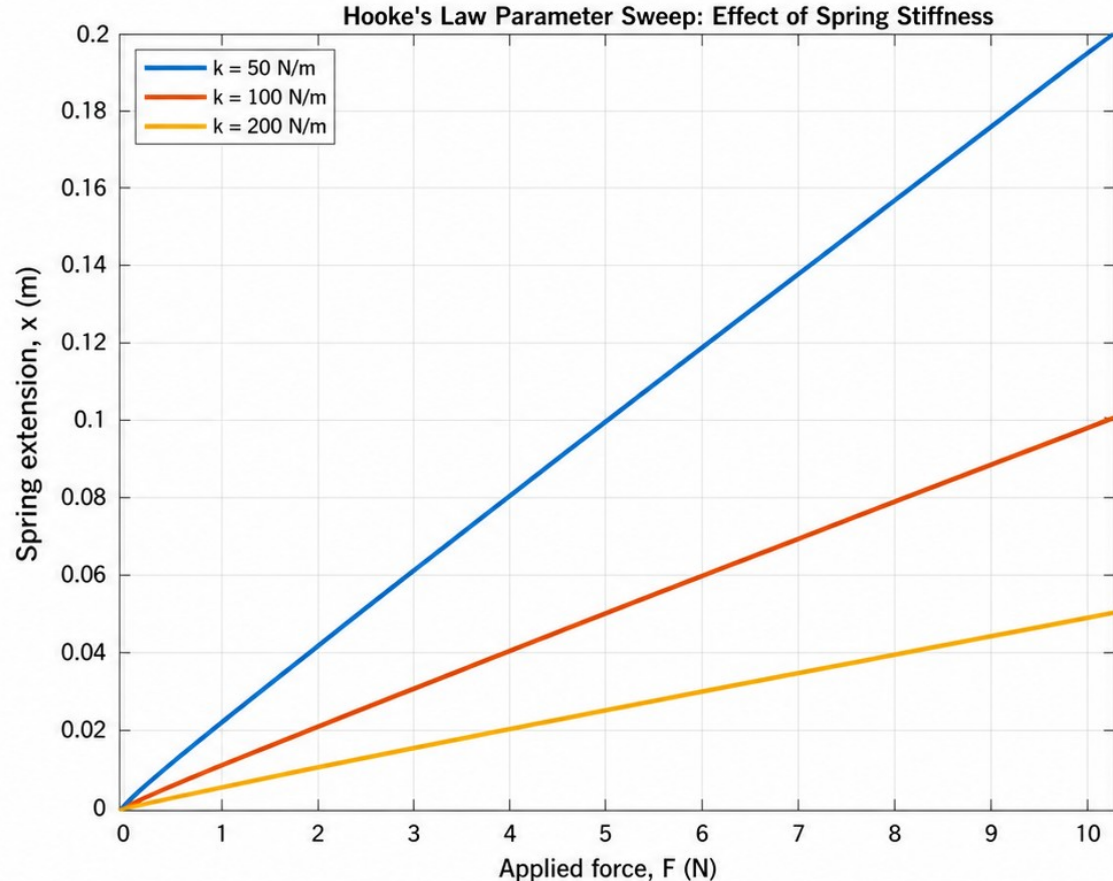
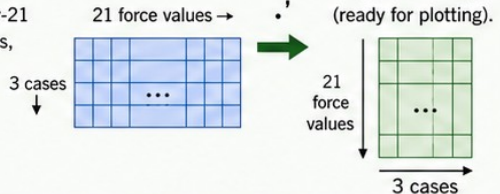
- ✓ Use one plot call for all stiffness cases; label the physical axes and identify each stiffness in the legend.
- ✓ The transpose changes the stored 3-by-21 arrangement into a plotting arrangement without changing the values.

## MATLAB plotting command

```
plot(F_N,extension_m.','LineWidth',2)
xlabel('Applied force, F (N)')
ylabel('Spring extension, x (m)')
legend('k = 50 N/m', 'k = 100 N/m', 'k = 200 N/m', ...
       'Location','northwest')
```

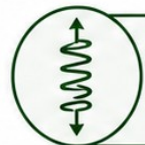
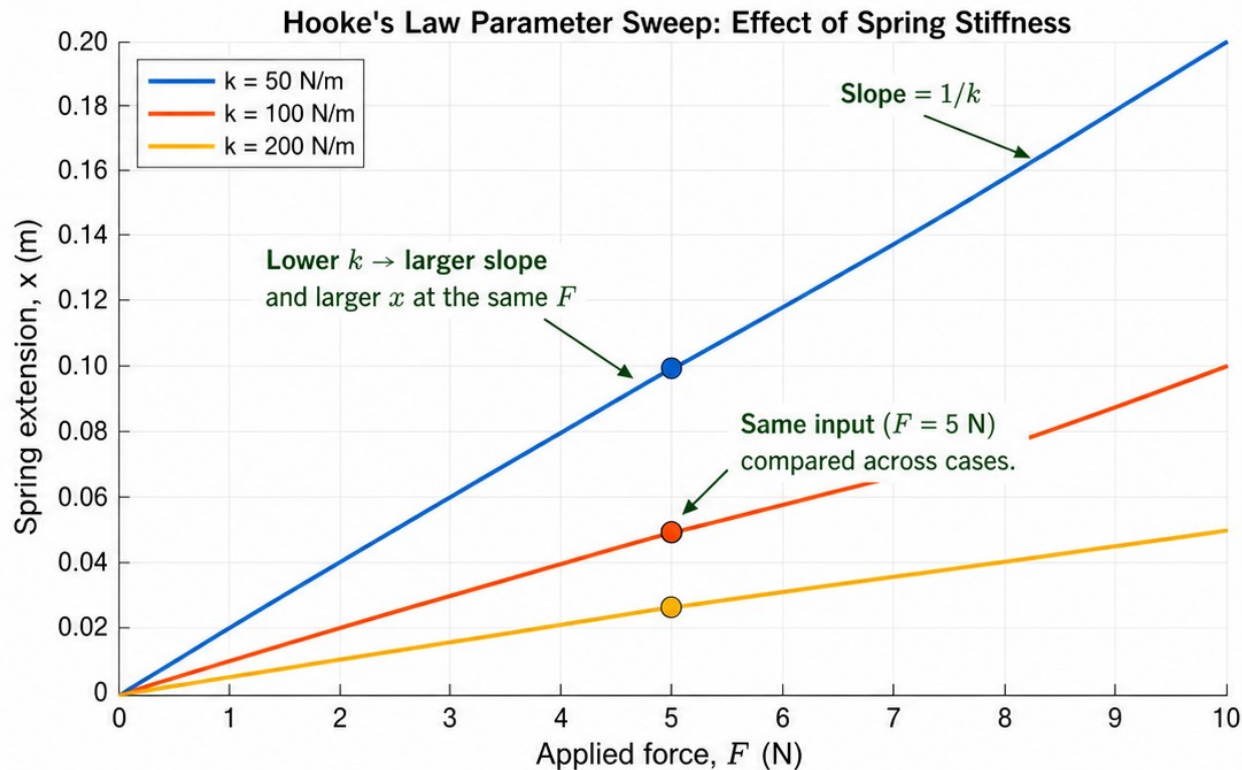
## Transpose operation ( '.')

Stored result array:  
extension\_m is 3-by-21  
(three stiffness cases,  
21 force values).



# The graph is a physics statement

- ✓ Each curve is straight because  $x$  is proportional to  $F$ .
- ✓ The slope is  $1/k$ .
- ✓ Smaller  $k$  gives a steeper line and larger extension at the same force.
- ✓ The graph supports the prediction only because the comparison uses common force inputs.



## Physical interpretation:

A softer spring (smaller  $k$ ) stretches more under the same force.  
A stiffer spring (larger  $k$ ) resists more and extends less.

$$x = \frac{F}{k}$$

# Validate a known limiting case

- ✓ At  $F = 0$ , Hooke's law requires  $x = 0$  for every positive stiffness.
- ✓ Read the first result column directly.
- ✓ Assert that the largest absolute value is below  $1\text{e-}12$ .
- ✓ A known limit turns a physical expectation into a reproducible check.

## MATLAB validation code (strict)

```
zero_force_extension_m = extension_m(:,1)
assert(max(abs(zero_force_extension_m)) < 1e-12, ...
       'Zero-force limiting-case validation failed.')
```

**What this tests:** The evaluated model output across all cases.

**Not a syntax check:** It confirms the physics at a known limit.

## Zero-force result (first column of extension\_m)

At  $F = 0$  N,  
the spring extension  
for every stiffness is  
exactly zero.

$$\text{extension\_m}(:,1) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ m}$$


$\max(\text{abs}(\text{extension\_m}(:,1))) = 0 < 1\text{e-}12$

**Zero-force limiting-case validation passed.**

**?** **Think:** Why must all three values be zero? What would a nonzero value indicate?



# A runnable calculation can still be wrong



## DIAGNOSIS: MATLAB runs, but the physics is wrong

### Wrong vs correct expressions

```
wrong_baseline_extension = F_N.*baseline_k_Npm;  
correct_baseline_extension = F_N./baseline_k_Npm;  
wrong_final_value = wrong_baseline_extension(end)  
correct_final_value_m = correct_baseline_extension(end)
```

### Model being used (unchanged)

$$F = kx, \quad x = \frac{F}{k}$$

### Inputs for all cases (fixed)

- Applied force:  $F, N = 0:0.5:10 \text{ N}$
- Stiffness array:  $k_{\text{Npm}} = [50 \ 100 \ 200] \text{ N/m}$
- Baseline stiffness:  $\text{baseline\_k\_Npm} = 100 \text{ N/m}$



## PHYSICAL CHECK: The wrong expression fails

Faulty expression:  $x = F.*k$



### Unit mismatch

$F$  is in N and  $k$  is in N/m.

$N \times (N/m) = N^2/m$ ,  
not metres (m).



### Reversed trend

At the same force, larger  $k$  gives  
**larger**  $x$  in the wrong model.

Correct model  $x = F/k$

$k \uparrow \Rightarrow x \downarrow$   
(stiffer  $\rightarrow$  smaller extension)

Wrong model  $x = F.*k$

$k \uparrow \Rightarrow x \uparrow$   
(stiffer  $\rightarrow$  larger extension)

# Repair the expression, then recheck the physics

- ✓ At  $F = 10$  N, compare correct  $x = F/k$  with wrong  $x = F.*k$ .
- ✓ The correct model gives 0.20, 0.10, and 0.05 m; the wrong expression gives 500, 1000, and 2000 N<sup>2</sup>/m.
- ✓ Repair by tracing the model, units, limiting case, and expected trend.

Comparison at  $F = 10$  N

| $k$ (N/m) | Correct $x = F/k$ (m) | Wrong $x = F.*k$ (N <sup>2</sup> /m) |
|-----------|-----------------------|--------------------------------------|
| 50        | 0.20                  | 500                                  |
| 100       | 0.10                  | 1000                                 |
| 200       | 0.05                  | 2000                                 |

## Repair path: from suspicion to correction

1

$$F = kx, \\ x = \frac{F}{k}$$

Trace the model

Hooke's law:  $F = kx$ ,  $x = \frac{F}{k}$ .

2

$$[k] = \text{N/m}$$

Check units

$x$  must be in m.  
The wrong output has units N<sup>2</sup>/m.

3

$$F = 0 \\ \Rightarrow x = 0$$

Apply the limiting case

At  $F = 0$ , every case must give  $x = 0$ .

4

$$x \propto \frac{1}{k}$$

Check the expected trend

Larger  $k$  should give smaller  $x$ .  
The wrong expression gives the opposite.

5



Correct the expression

Use element-wise division for  $x = F/k$ .

```
wrong_baseline_extension = F_N.*baseline_k_Npm;  
correct_baseline_extension = F_N./baseline_k_Npm;  
wrong_final_value = wrong_baseline_extension(end)  
correct_final_value_m = correct_baseline_extension(end)
```

```
correct_baseline_extension = F_N./baseline_k_Npm;
```



**Recheck the physics:** at  $F = 0$ , every case gives  $x = 0$ ; at  $F = 10$  N, the extensions are 0.20, 0.10, and 0.05 m.



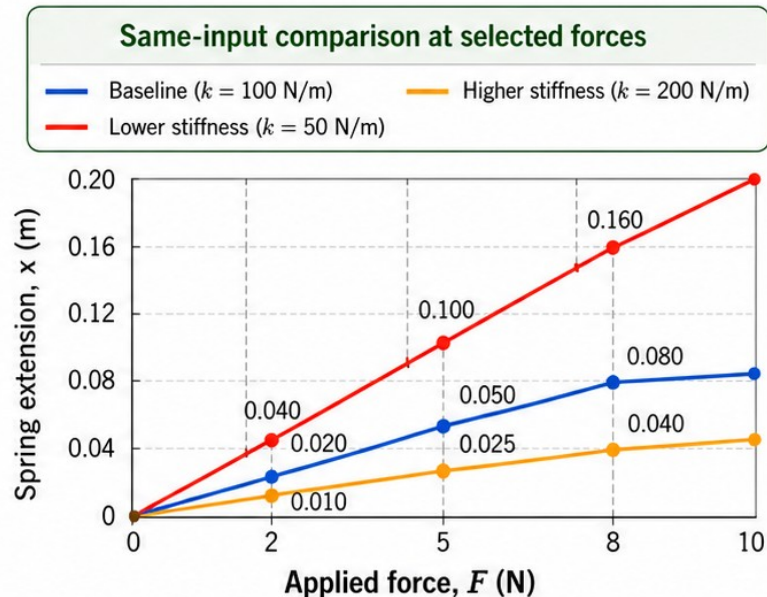
# Working exposure: compare selected inputs

- ✓ Compare the same selected forces for every stiffness case.

- ✓ At  $F = [2 \ 5 \ 8] \text{ N}$ , the values follow the same inverse-stiffness ordering.

- ✓ A table and a graph are two views of the same controlled experiment.

| $F \text{ (N)}$ | $k = 50 \text{ N/m}$ | $k = 100 \text{ N/m}$ | $k = 200 \text{ N/m}$ |
|-----------------|----------------------|-----------------------|-----------------------|
| 2               | 0.040                | 0.020                 | 0.010                 |
| 5               | 0.100                | 0.050                 | 0.025                 |
| 8               | 0.160                | 0.080                 | 0.040                 |



- At the same  $F$ , extensions follow the extension order:  
 $x_{50} > x_{100} > x_{200}$ .



# Exit ticket — explain a fair sweep to a future you



## PREDICT

the trend from the equation.

### Hooke's law

$$F = kx \text{ and } x = \frac{F}{k}$$

For a fixed  $F > 0$ :

$$k \uparrow \longrightarrow x \downarrow$$

Stiffer spring  $\rightarrow$  smaller extension.



## CONTROL

one changed parameter and the fixed inputs.

### Swept parameter:

spring stiffness,  $k$  (N/m)

$$k_{Npm} = [50 \ 100 \ 200] \text{ N/m}$$

### Fixed model and input array:

Ideal Hooke's law,  $x = F/k$

$$F_N = 0:0.5:10 \text{ N}$$

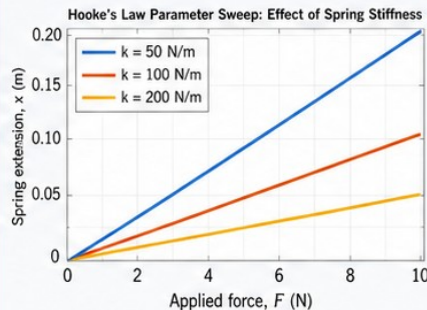
Same model, same  $F$ , different  $k$ .



## PLOT

labelled cases on common axes.

### Common axes:



Compare cases at the same  $F$ .



## CHECK & INTERPRET

a known or limiting case, then the physical result.

### Known or limiting case:

At  $F = 0$  N, all cases give  $x = 0$  m. (Passes.)

### Interpretation:

For the same force, a larger  $k$  produces a smaller extension. Matches  $x = F/k$ .

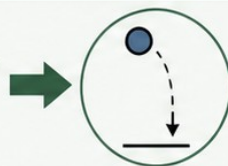


## Transfer the workflow

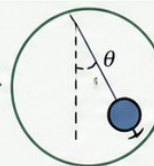
to analogous supplied models.



Ohmic resistor  
 $V = IR$



vertical motion  
 $s = ut + \frac{1}{2}at^2$



small-angle  
pendulum

$$T = 2\pi\sqrt{L/g}$$